

Correlation between migration background and education level

Tamara Bakker 2695715, Rik de Bruijn 2894496, Joep Hagendoorn 2893599, Lize den Hertog 2889649, Isabelle Stankovski

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Set-up your environment

```
## Loading required package: tidyverse

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.3      v tibble    3.3.1
## v lubridate   1.9.5      v tidyr     1.3.2
## v purrr       1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Loading required package: rmarkdown
##
## Loading required package: yaml
##
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

Title Page

Tamara Bakker, Rik de Bruijn, Joep Hagendoorn, Lize den Hertog and Isabelle Stankovski

Tutorial group 2-1

Fatima Amankour

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

A study by research agency SCALIQ involving 6409 secondary school students, suggests that students with Arabic first and/or last names are more likely to be placed at a lower educational level than equally intelligent student with European sounding names. About half of the students with Arabic names were placed half a school level lower. The researchers say that this may be a sign of inequality in education. However, the

study only looks at intelligence and names. It did not look at other factors such as family income, home support or cultural background. The experts say that being placed in a lower school level that is too low can hurt students' education, confidence/motivation and career opportunities.

Part 2 - Data Sourcing

2.1 Load in the data

2.2 Provide a short summary of the dataset(s)

Summary dataset on education level and migration background

In the first dataset about education level and migration background you can see that a group consisting of both men and women was researched. All of the men and women are 15 years old and above. In the dataset you can see the different migration backgrounds with the highest level of education achieved.

```
##          ID          Geslacht      Leeftijd      Migratieachtergrond
## Min.      : 190      Length :12474      Min.      :20150      Length :12474
## 1st Qu.: 41410      N.unique :    1      1st Qu.:52052      N.unique :   11
## Median : 83003      N.blank  :    0      Median :53500      N.blank  :    0
## Mean      : 82923      Min.nchar:    7      Mean      :46186      Min.nchar:    7
## 3rd Qu.:124316      Max.nchar:    7      3rd Qu.:53800      Max.nchar:    7
## Max.      :165536                                Max.      :53925
## HoogstBehaaldOnderwijsniveau      Perioden      Bevolking_1
## Min.      :2018710                                Length :12474      Min.      : 0.0
## 1st Qu.:2018720                                N.unique :   18      1st Qu.:  2.0
## Median :2018770                                N.blank  :    0      Median : 16.0
## Mean      :2018764                                Min.nchar:    8      Mean      :130.7
## 3rd Qu.:2018800                                Max.nchar:    8      3rd Qu.:  77.0
## Max.      :2018810                                Max.      :3469.0
```

Summary dataset on migration background percentage per region

```
## Provincienaam % Herkomst Nederland % Herkomst buiten Nederland
## Length :12      Min.      :0.6100      Min.      :0.1300
## N.unique :12      1st Qu.:0.6950      1st Qu.:0.1900
## N.blank  : 0      Median :0.7650      Median :0.2350
## Min.nchar: 7      Mean      :0.7475      Mean      :0.2525
## Max.nchar:13      3rd Qu.:0.8100      3rd Qu.:0.3050
## Max.      :0.8700      Max.      :0.3900
## % Herkomst Europa % Herkomst buiten Europa
## Min.      :0.05000      Min.      :0.0800
## 1st Qu.:0.06750      1st Qu.:0.1150
## Median :0.08500      Median :0.1300
## Mean      :0.08917      Mean      :0.1633
## 3rd Qu.:0.11250      3rd Qu.:0.2175
## Max.      :0.14000      Max.      :0.3000
## % Geboren in Nederland met herkomst Nederland
## Min.      :0.6100
## 1st Qu.:0.6950
## Median :0.7650
```

```

## Mean      :0.7475
## 3rd Qu.:0.8100
## Max.      :0.8700
## % Geboren in Nederland met herkomst Europa
## Min.      :0.0200
## 1st Qu.:0.0300
## Median :0.0300
## Mean      :0.0325
## 3rd Qu.:0.0325
## Max.      :0.0600
## % Geboren in Nederland met herkomst buiten Nederland
## Min.      :0.050
## 1st Qu.:0.080
## Median :0.095
## Mean      :0.105
## 3rd Qu.:0.135
## Max.      :0.170
## % Geboren in Nederland met herkomst buiten Europa % Geboren in Nederland
## Min.      :0.0300      Min.      :0.7700
## 1st Qu.:0.0475      1st Qu.:0.8325
## Median :0.0550      Median :0.8500
## Mean      :0.0700      Mean      :0.8517
## 3rd Qu.:0.1025      3rd Qu.:0.8900
## Max.      :0.1400      Max.      :0.9200
## % Geboren buiten Nederland met herkomst Europa
## Min.      :0.03000
## 1st Qu.:0.04000
## Median :0.05500
## Mean      :0.05583
## 3rd Qu.:0.07250
## Max.      :0.09000
## % Geboren buiten Nederland met herkomst buiten Europa
## Min.      :0.05000
## 1st Qu.:0.06750
## Median :0.07500
## Mean      :0.09167
## 3rd Qu.:0.12000
## Max.      :0.16000
## % Geboren buiten Nederland
## Min.      :0.0800
## 1st Qu.:0.1100
## Median :0.1500
## Mean      :0.1483
## 3rd Qu.:0.1675
## Max.      :0.2300

```

Summary dataset on education level per region

```

##      Regio's      Perioden
## Length      :40      Min.      :2023
## N.unique :14      1st Qu.:2023
## N.blank  : 0      Median :2024
## Min.nchar: 9      Mean      :2024

```

```
## Max.nchar:18      3rd Qu.:2025
##                  Max.    :2025
##                  NAs      :1
## Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Basisonderwijs, vmbo, mbo1 (%)
## Min.    :22.30
## 1st Qu.:25.93
## Median :26.55
## Mean    :26.18
## 3rd Qu.:27.27
## Max.    :28.80
## NAs     :14
## Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Havo, vwo, mbo2-4 (%)
## Min.    :35.50
## 1st Qu.:40.83
## Median :42.70
## Mean    :42.51
## 3rd Qu.:44.92
## Max.    :46.80
## NAs     :14
## Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Hbo, wo (%)
## Min.    :24.90
## 1st Qu.:27.75
## Median :31.75
## Mean    :31.32
## 3rd Qu.:33.10
## Max.    :42.20
## NAs     :14
```

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Renaming unclear characters in edu_mig_dataset

Eliminating unnecessary observations and variables in edu_mig_dataset

Renaming inconsistent characters in reg_edu_dataset

Eliminating unnecessary years in reg_edu_dataset

Eliminating unnecessary variables in reg_mig_dataset

3.2 Generate necessary variables

Variable 1: Percentage of the total of Dutch people and non-European migrants in each level of education

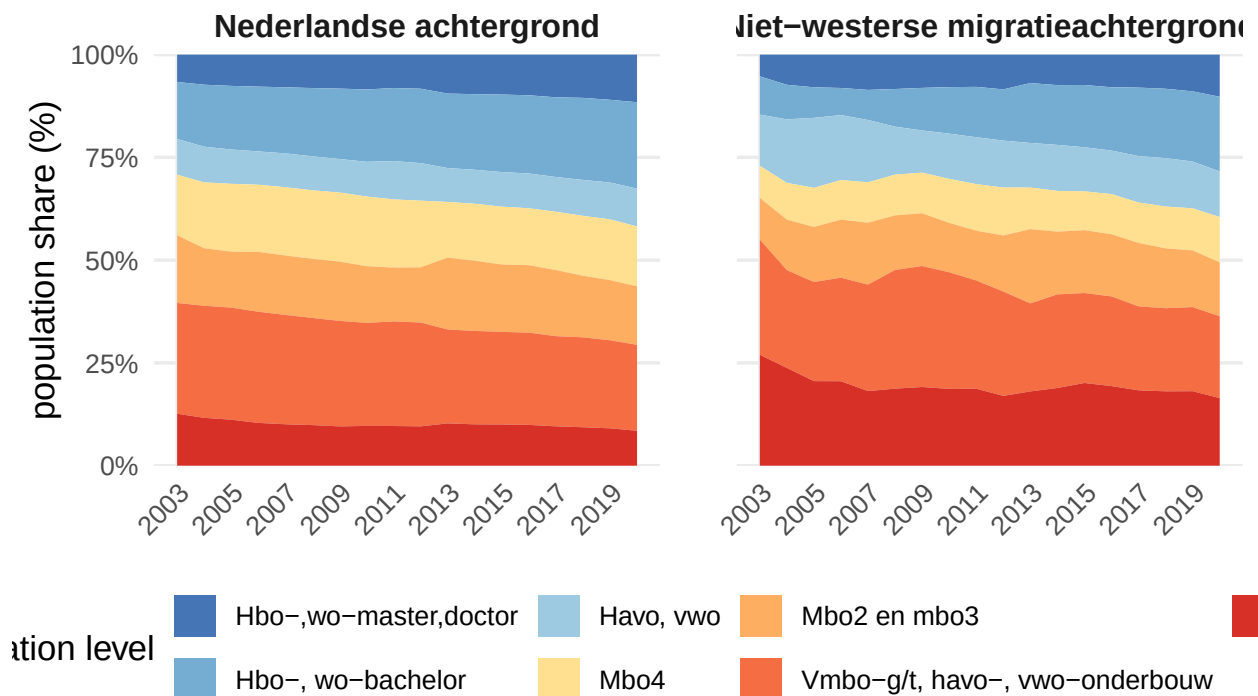
Variable 2: Percentage of non-European migrants that have obtained a hbo or wo bachelor for different age groups in 2020

3.3 Visualize temporal variation

Make own variable that shows our relation and plot it over time. Line graph with other version of migration-education dataset.

Highest level of education based on migration background

Population 15 year of older, 2003–2020 | source: CBS (EBB, tabel 82275)



3.4 Visualize spatial variation

Compare/merge two maps, one with the level of education of a province and one with the proportion of migrants/people with a migration background living there. use datasets on migration background per region and education level per region to make maps. requires separate package, preferably from CBS. test effe

```
# Kolomnamen opslaan
basis_col <- "Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Basisonderwijs, vmbo, mbo1 (%)"

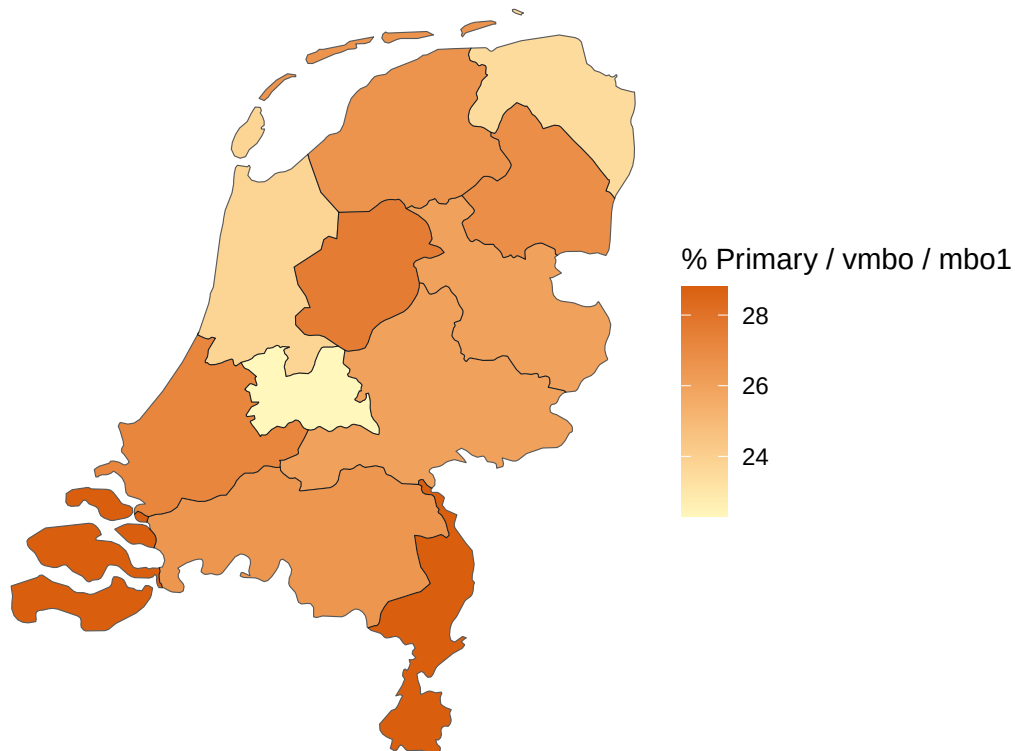
nl_map_data <- ne_countries(country = "Netherlands", returnclass = "sf")
nl_provinces <- ne_states(country = "Netherlands", returnclass = "sf")

# Herstel namen naar Nederlands
nl_provinces$name <- recode(nl_provinces$name,
  "North Brabant" = "Noord-Brabant",
  "South Holland" = "Zuid-Holland",
  "North Holland" = "Noord-Holland"
)

# Join
spatial_var_edu <- right_join(nl_provinces, reg_edu_dataset_cl, by = c("name" = "Regio's"))

# Plot
ggplot(spatial_var_edu) +
  geom_sf(aes(fill = .data[[basis_col]])) +
  scale_fill_gradient(low = "#fff7bc", high = "#d95f0e") +
  labs(
    title = "Percentage of population with primary education or lower (per province)",
    fill = "% Primary / vmbo / mbo1"
  ) +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

ge of population with primary education or lower (per province)



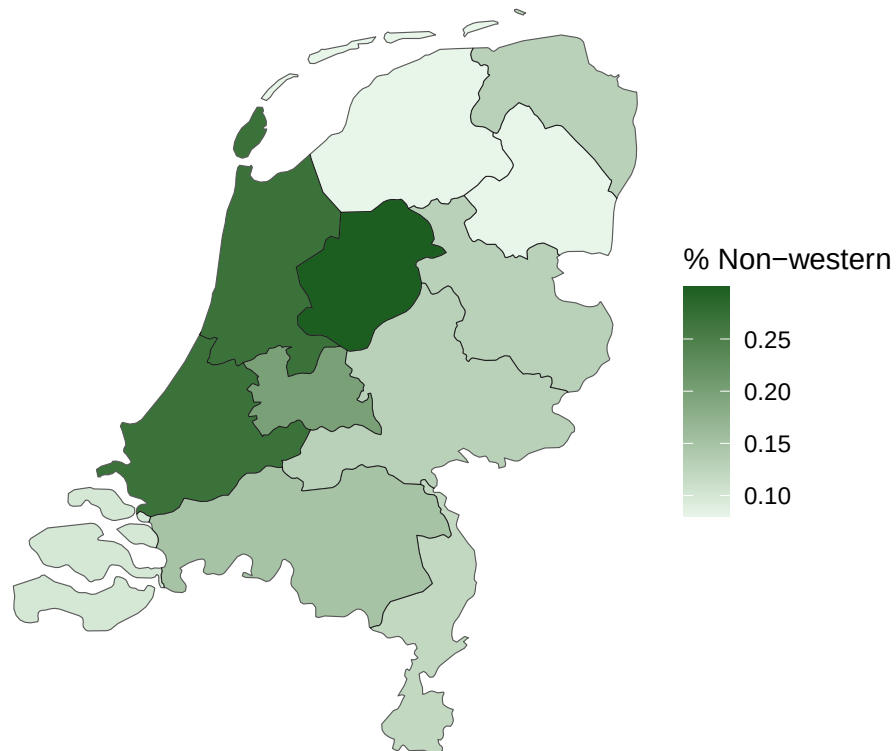
```
# Laad kaartdata
nl_map_data <- ne_countries(country = "Netherlands", returnclass = "sf")
nl_provinces <- ne_states(country = "Netherlands", returnclass = "sf")

# Herstel namen naar Nederlands
nl_provinces$name <- recode(nl_provinces$name,
  "North Brabant" = "Noord-Brabant",
  "South Holland" = "Zuid-Holland",
  "North Holland" = "Noord-Holland"
)

# Join
spatial_var_mig <- right_join(nl_provinces, reg_mig_dataset_cl, by = c("name" = "Provincienaam"))

# Plot
ggplot(spatial_var_mig) +
  geom_sf(aes(fill = `% Herkomst buiten Europa`)) +
  scale_fill_gradient(low = "#e8f5e9", high = "#1b5e20") +
  labs(
    title = "Percentage of population with non-Western migration background",
    fill = "% Non-western"
  ) +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

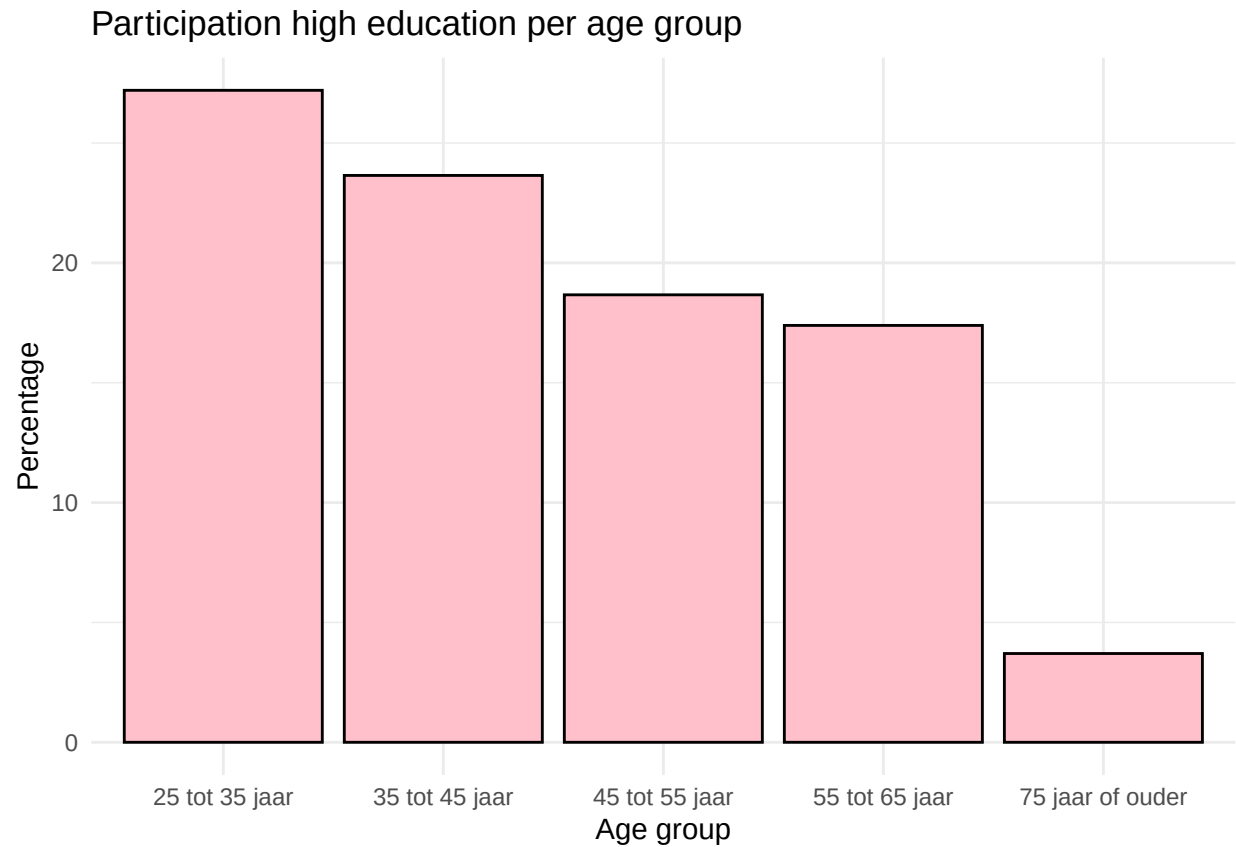
Percentage of population with non-Western migration background



Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

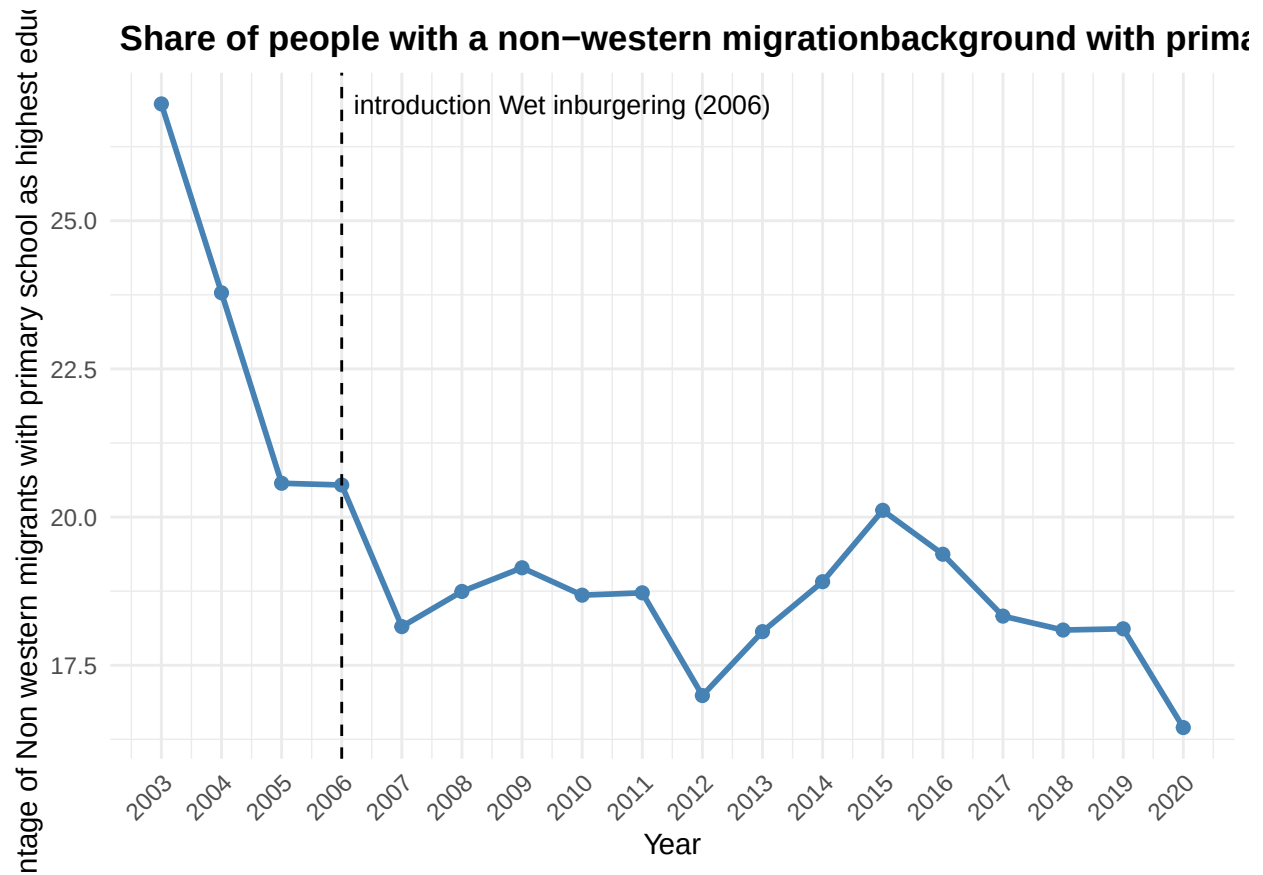
Comparison between the difference in education level between different ages, use dataset migration education and either a histogram or bar plot.



Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Find some kind of event in order to show changes caused by this event: Analyze the relationship between two variables. will be a line graph, not sure which dataset yet.



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Discuss the findings from all the analyses and draw somewhat of a conclusion whether our hypothesis was right. # Part 5 - Reproducibility make sure that if someone were to reproduce our work, a change in packages would not matter as the information of which packages we used when is clear. ## 5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.